Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

(Signature and Date in Handwritten)

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Session	**2026-27 (Odd)**	**Course Name**	**LAB: Optimization technique**
Semester	**7th**	**Course Code**	**23ADS1708**
Roll No	**39**	**Name of Student**	**Hrutik Gujarkar**

Practical Number	**9**
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To solve integer programming problems using the Branch and Bound Method.
Problem Definition	To maximize $Z = 5x_1 + 4x_2$ subject to $x_1 + x_2 \leq 5$, $10x_1 + 6x_2 \leq 45$ and $x_1, x_2 \geq 0$ and integers, using the Branch and Bound Method.
Theory (100 words)	In an integer programming problem, some or all variables must take integer values. The Branch and Bound method first solves the LP relaxation, which ignores the integer restrictions. If a variable x_j has a fractional value v, the problem is divided (branched) into two subproblems: $x_j \leq$ floor(v) and $x_j \geq$ ceil(v). For a maximization problem, the LP value of a node is an upper bound for all integer solutions in that branch. A node is pruned if it is infeasible, gives an integer solution, or its bound is not better than the best integer value found so far. The search ends when no active node remains.
	Step-by-Step Method 1. Solve the LP relaxation of the problem by ignoring the integer restrictions. If the solution is an integer, stop.

	2. Otherwise, select a variable x_j having a fractional value v. 3. Create two subproblems by adding the constraints $x_j \leq$ floor(v) and $x_j \geq$ ceil(v). 4. Solve the LP of each subproblem. 5. Prune a node if it is infeasible, if it gives an integer solution (update the best value), or if its Z is not better than the best integer Z. 6. Branch further on the remaining nodes whose Z is better than the best integer Z. 7. When no active node remains, the best integer solution is the optimum.
Procedure and Execution (100 Words)	Algorithm: 1. Define the objective function and the constraints. 2. Solve the LP relaxation at the current node. 3. If the node is infeasible, prune it. 4. If Z is not better than the best integer value, prune it. 5. If all variables are integers, update the best solution. 6. Otherwise, choose a fractional variable and create two branches. 7. Repeat for each branch (recursively). 8. Display the optimal integer solution and the maximum value of Z.

	Code:
	`import math`
	`from scipy.optimize import linprog`
	
	`# Maximize Z = 5*x1 + 4*x2`
	`# subject to  x1 + x2 <= 5`
	`#              10*x1 + 6*x2 <= 45`
	`#              x1, x2 >= 0 and integer`
	`c = [-5, -4]           # negated because linprog minimizes`
	`A = [[1, 1], [10, 6]]`
	`b = [5, 45]`
	
	`best_value = -math.inf`
	`best_sol = None`
	`node_count = 0`
	
	`def branch_and_bound(bounds, parent, cond):`
	`  global best_value, best_sol, node_count`
	`  node_count += 1`
	`  node = node_count`
	`  if parent is None:`
	`  label = f"Node {node} (root LP)"`
	`  else:`
	`  label = f"Node {node} (from Node {parent}, {cond})"`
	`  res = linprog(c, A_ub=A, b_ub=b, bounds=bounds, method="highs")`
	`  if not res.success:`

```
        print(f"{label}: infeasible -> pruned")
        return
    z = -res.fun
    x1, x2 = res.x
    print(f"{label}: x1 = {x1:.3f}, x2 = {x2:.3f}, Z = {z:.3f}")
    if z <= best_value + 1e-9:
        print("   Bound <= best integer value -> pruned")
        return
    # Find a fractional variable
    frac = [i for i, v in enumerate(res.x)
            if abs(v - round(v)) > 1e-6]
    if not frac:
        print("   Integer solution found -> new best")
        best_value, best_sol = z, [round(v) for v in res.x]
        return
    i = frac[0]
    v = res.x[i]
    name = f"x{i+1}"
    lo, hi = bounds[i]
    # Left branch: xi <= floor(v)
    left = list(bounds); left[i] = (lo, math.floor(v))
    branch_and_bound(left, node, f"{name} <= {math.floor(v)}")
    # Right branch: xi >= ceil(v)
    right = list(bounds); right[i] = (math.ceil(v), hi)
    branch_and_bound(right, node, f"{name} >= {math.ceil(v)}")

print("Branch and Bound Tree:")
branch_and_bound([(0, None), (0, None)], None, "")

print("\nTotal LP problems solved:", node_count)
print("Optimal Integer Solution: x1 = %d, x2 = %d" % tuple(best_sol))
print("Maximum Value of Z:", round(best_value))
```

Output:

```
Branch and Bound Tree:
Node 1 (root LP): x1 = 3.750, x2 = 1.250, Z = 23.750
Node 2 (from Node 1, x1 <= 3): x1 = 3.000, x2 = 2.000, Z = 23.000
   Integer solution found -> new best
Node 3 (from Node 1, x1 >= 4): x1 = 4.000, x2 = 0.833, Z = 23.333
Node 4 (from Node 3, x2 <= 0): x1 = 4.500, x2 = 0.000, Z = 22.500
   Bound <= best integer value -> pruned
Node 5 (from Node 3, x2 >= 1): infeasible -> pruned

Total LP problems solved: 5
Optimal Integer Solution: x1 = 3, x2 = 2
Maximum Value of Z: 23
```

Output Analysis	The LP relaxation at the root node (Node 1) gave $x_1 = 3.75$, $x_2 = 1.25$ and $Z = 23.75$, which is an upper bound. Branching on x_1 gave Node 2 ($x_1 \leq 3$) with the integer solution (3, 2) and $Z = 23$, which became the best integer solution. Node 3 ($x_1 \geq 4$) gave $Z = 23.333$ with a fractional x_2, so it was branched on x_2. Node 4 ($x_2 \leq 0$) gave $Z = 22.5$, which is less than 23, so it was pruned. Node 5 ($x_2 \geq 1$) was infeasible. Hence the optimum is $x_1 = 3$, $x_2 = 2$ with $Z = 23$, found after solving 5 LP problems.
Link of student Github profile where lab assignment has been uploaded	https://github.com/HrutikGujarkar/Optimizaton-techniques-Practicals
Conclusion	The integer programming problem was successfully solved using the **Branch and Bound Method**. The optimal integer solution is $x_1 = 3$ and $x_2 = 2$, and the maximum value of Z is **23**.
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